

- A domino covers $(i+1, j)$ and $(i+2, j)$.
- A domino covers $(i+3, j)$ and $(i+4, j)$.
- This process is continued until a domino covers $(i-2, j)$ and $(i-1, j)$.

Each of the 7 columns apart from j is covered by dominoes as follows:

- A Domino covers Row 1 and Row 2 of that column.
- Another Domino covers Row 3 and Row 4 of that column.
- Yet Another Domino covers Row 5 and Row 6 of that column.
- The Last Domino covers Row 7 and Row 8 of that column. $\square$

49. Show that by removing two white squares and two black squares from an 8x8 checkerboard (colored as in the text) you can make it impossible to tile the remaining squares using dominoes. (Very Good Problem)

Solution:

of Row 1 column
by dominoes in
with Row 1 column

- Remove the 1 column 7 and 1 column 8.
- The white squares are

W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W

The Black square 8 can only be covered 2 ways: Either 7 and Row 2 column 8. White squares of Row 2 column 8:

- Remove the 1 column 7 and 1 column 8.
- The white squares are
- 2 options: The Black squares of Row 7 column 8 and Row 8 column 7.
- Remove the Black squares of Row 7 column 8 and Row 8 column 7.
- It's easy to see that the rest of the checkerboard cannot be covered by dominoes.

• We have given an example of an 8x8 checkerboard where it is impossible to cover it using dominoes if we remove two white and two black squares. $\square$

51. a) Draw each of the five different tetrominoes, where a tetromino is a polyomino consisting of four squares.

Solution:

Tetromino 1:

W	B	W	B
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Tetromino 2:

W	B	W	B
		B	